NARAYANA IIT ACADEMY INDIA

JEE ADVANCED GRAND TEST SERIES



46 YEARS OF EXCELLENCE

NARAYANA IIT - JEE ACADEMY - INDIA

Sec: OSR.IIT _*CO-SC(MODEL-A&APEX) **Date: 29-03-26**
Time: 3 Hrs **Max. Marks: 198**

29-03-26_OSR.IIT_*CO-SC(MODEL-A&APEX)_JEE ADV_GTA-2(P2)_SYLLABUS

Rankers Academy JEE

MATHEMATICS: **TOTAL SYLLABUS**

PHYSICS: **TOTAL SYLLABUS**

CHEMISTRY: **TOTAL SYLLABUS**

THE NARAYANA GROUP

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TIME: 3HRS **IMPORTANT INSTRUCTIONS** **Max Marks: 198**

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 6)	Questions with Single digit integer(0-9)	+3	-1	6	18
Sec – II (Q.N : 7 – 12)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 13 – 18)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -1 27.30)	+4	0	6	24
Total				**18**	**66**

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 19 – 24)	Questions with Single digit integer(0-9)	+3	-1	6	18
Sec – II (Q.N : 25 – 30)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 31 – 36)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -1 27.30)	+4	0	6	24
Total				**18**	**66**

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 37 – 42)	Questions with Single digit integer(0-9)	+3	-1	6	18
Sec – II (Q.N : 43 – 48)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 49 – 54)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -1 27.30)	+4	0	6	24
Total				**18**	**66**

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MATHEMATICS **MAX.MARKS: 66**

SECTION – 1 (Maximum Marks: 18)

This section contains SIX (06) questions.
The answer to each question is **A SINGLE DIGIT INTEGER** rangeing from **0 TO 9 , BOTH INCLUSIVE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric
keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

1. Let z_1 and z_2 be two complex number satisfying the following system of equation:

$z_1^3 - 3z_1z_2^2 = 2$

$3z_1^2z_2 - z_2^3 = 11$

Find the value of $\left|z_1^2 + z_2^2\right|$.

2. Let P be the probability such that the sum of the outcomes of rolling 10 fair dice is divisible by 7. Then the value of $[1/p]$ (where [.] is G.I.F)

3. A variable line $ax+by+c=0$. where the coefficients a,b and c are in arithmetic progression (A.P), is a normal to the circle defined by the equation:

$(x-\alpha)^2+(y-\beta)^2=\gamma$ This circle is known to be orthogonal to the circle.

$x^2+y^2-4x-4y-1=0$. Then the numerical value of $(\alpha+\beta+\gamma)$ is

4. Let A be a 3×3 matrix such that $tr(A)=3, tr(A^2)=5$, and $tr(A^3)=15$. Find the value of $\det(A^2-3A+2I)$.

5. Let α denoted the number of points of discontinuity of the function:

$$f(x)=|x|+\left|\frac{x}{3}-1\right|+\left||x|+\left|\frac{x}{3}-1\right|\right|$$

Let β denote the sum of all the roots of the equation $g(x)-h(x)=0$ in the interval $[-2,2]$, where:

$$g(x)=\left|2-|x-2|\right| \text{ and } h(x)=\frac{|\sin x|}{\sin x}(x\neq0)$$

Find the value of $\alpha+\beta$

6. Let $L=\lim\limits_{x\to0^+}\dfrac{e^{\left(x^x-1\right)}-x^x}{\left(\left(x^2\right)^x-1\right)^2}$ value of $\dfrac{1}{L}$ is.

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SECTION - 2 (Maximum Marks : 24)

This section contains SIX (06) questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

7. Let $f:[-1,2]\to\mathbb{R}$ be a function defined by:

$$f(x)=|x[x]|+x\left[\cos\left(\frac{\pi x}{2}\right)\right]$$

(A function $f(x)$ is said to have a jump discontinuity at a point $x=c$ if the one – sided limits at that exist as finite value but are not equal to each other)

(Where $[.]$ denotes the greatest integer function) then which of the following statements is/ are TRUE

A) $f(x)$ is continuous at $x=0$ but not differentiable at $x=0$.

B) $f(x)$ has a jump discontinuity at $x=1$.

C) $f(x)$ has a jump discontinuity at $x=2$.

D) $f(x)$ is differentiable for all $x\in(1,2)$.

8. Which of the following option is / are correct about conic C, where

$C\equiv 9x^2+16y^2+24xy-46x-28y-3=0$

A) $(7-4)$ lies on axis

B) $25x+25y=28$ passes through vertex

C) $(-2,0)$ lies on directrix

D) $25x+28y=27$ passes through focus

9. Let $g:R\to R$ be a differentiable function defined by:

$\int\cos^x(x)\left(\ln(\cos x)-x\tan x\right)dx=g(x)+c$ where $\alpha=\dfrac{g'\left(\frac{\pi}{3}\right)}{-\left(\frac{1}{2}\right)^{\pi/3}}$ and $\dfrac{e^{\alpha}}{q}=e^{\frac{\pi}{\sqrt{p}}}$, where p, q are natural numbers. Then which of the following is/ are correct.

A) $p+q$ is prime number

B) p is prime but q is not prime

C) p and q both prime numbers

D) $\log_q^p\ \log_p^{(2q-1)}=0$

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10. Consider the line $L:\frac{x-1}{2}=\frac{y-2}{1}=\frac{z-3}{-2}$ and the plane $P:x+y+z-3=0$.

Then which of following statements: is / are correct.

A) The images of point (1,2,3) with respect to plane P is $(-1,0,1)$

B) Line L and plane P intersect each other.

C) The projection of line L on plane P is $\frac{x}{5}=\frac{y-1}{2}=\frac{z-1}{-7}$.

D) The image of line L with respect to plane P is $\frac{x+1}{4}=\frac{y}{-1}=\frac{z-1}{-8}$.

11. Let M,m denotes respectively maximum and minimum value of $\left|z^2-2iz+1\right|$(where $|z|=3$). Then which of the following is /are correct.

A) $M+m=18$ B) $M+m=16$ C) $M-m=14$ D) $M-m=12$

12. Let $\{\vec{a},\vec{b},\vec{c}\}$ be a set of three non- coplanar vectors and let $\{\vec{p},\vec{q},\vec{r}\}$ be their reciprocal system, defined by $\vec{p}=\frac{\vec{b}\times\vec{c}}{\left[\vec{a}\,\vec{b}\,\vec{c}\right]},\vec{q}=\frac{\vec{c}\times\vec{a}}{\left[\vec{a}\,\vec{b}\,\vec{c}\right]},\vec{r}=\frac{\vec{a}\times\vec{b}}{\left[\vec{a}\,\vec{b}\,\vec{c}\right]}$

Let $\left[\vec{a}\,\vec{b}\,\vec{c}\right]=V\neq 0$

Which of the following statements are correct?

A) $\vec{p}\times\vec{q}+\vec{q}\times\vec{r}+\vec{r}\times\vec{p}=\frac{\vec{a}+\vec{b}+\vec{c}}{V}$

B) $\left[\left(\vec{a}\times\vec{b}\right)\times\left(\vec{b}\times\vec{c}\right)\right].\vec{p}=V\left(\vec{b}.\vec{p}\right)$

C) $\vec{a}\times\left(\vec{b}\times\vec{c}\right)+\vec{b}\times\left(\vec{c}\times\vec{a}\right)+\vec{c}\times\left(\vec{a}\times\vec{b}\right)=V\left(\vec{p}+\vec{q}+\vec{r}\right)$

D) $\vec{a}.\vec{p}+\vec{b}.\vec{q}+\vec{c}.\vec{r}=3$

SECTION - 3 (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
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Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

13. Let $S=\sum_{r=0}^{n}\frac{\binom{n}{r}\binom{n}{n-r}}{r+1}$. If $n=10$, the number of zeros at the end of the value $(11S)$.

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14. If $\int_{\frac{\pi}{2}}^{\pi} \frac{x^2}{\left(x^2-1\right)\sin 2x+2x\cos 2x}dx=\ln k$ then $\left[\frac{1}{k}\right]$ is equal to (where $[.]$ denotes the greatest integer function)

15. Ten individuals -2 Americans (A_1,A_2), 2 Russians (R_1,R_2), 2 Chinese (C_1,C_2), 2 Dubaians (D_1,D_2), and 2 Sentinelese Indians (In_1,In_2) are to be seated in a row of 10 labeled chairs.

The following constraints must be satisfied:

1. Fixed Ends: The two Americans $(A_1 and\ A_2)$ must occupy the two extreme seats (Seat 1 and Seat 10)

2. The Dubaian Block: The two Dubaians must sit together as a pair (D_1D_2). Furthermore, this pair must be placed directly between either the two Russians $(R-D_1D_2-R)$ or the two Chinese $(C_1-D_1D_2-C_2)$

3. The Political Constraint : One American (A_1) is a staunch " Nationalist" and refuses to sit next to any Russians (R). The other American (A_2) has no such restriction

4. No Restrictions: The 2 Chinese (if not part of the Dubaian block), The 2 Sentinelese Indians and the 2 Russian (if not part of the Dubaian block) have no other seating constraints.

16. If the value of $\log_8\left(2^{50}\binom{50}{0}+2^{49}\binom{51}{1}+2^{48}\binom{52}{2}+....2^{o}\binom{100}{50}\right)$ is equal to

Note:- $\binom{n}{r}$ represents the value of ${}^nC_r$

17. Let $f:\mathbb{Z}\to\mathbb{Z}$ be a non-constant function satisfying the functional equation:

$f(2a)+2f(b)=f(f(a+b))(\forall a,b\in\mathbb{Z})$

Given the condition, $f(2)=-8$, define the following sets of points in Cartesian plane $\mathbb{R}^2$:

- $S_1:\{(x,y)\}\in\mathbb{R}^2\,|\,y=f(x)\}$
- $S_2:\{(x,y)\}\in\mathbb{R}^2\,|\,y^2=32x\}$

Let the intersection $S_1\cap S_2$ consist of two points A and B and then the area of the triangle ΔOAB, where O the origin is $(0,0)$

18. If $p=\sum_{r=1}^{50}\left\{\sum_{k=1}^{r}\left(\frac{(-1)^{r-1}}{k}{}^{50}C_r\right)\right\}$ then the value of $100(p)$ equals

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PHYSICS **MAX.MARKS: 66**

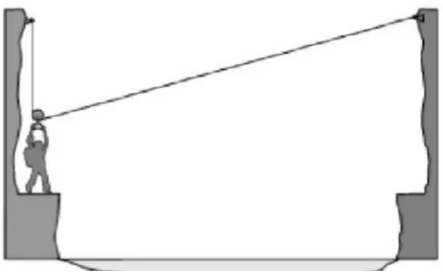
SECTION – 1 (Maximum Marks: 18)

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19. Consider an endothermic nuclear reaction $A+B\rightarrow C\left(Q=-\frac{E}{4}\right)$. Mass of nucleus A and B are m and 2m and they are moving with kinetic energies 2E and E respectively in the same direction before collision. Assume the collision to be perfectly inelastic, thus forming daughter nucleus C. C gets excited by absorbing all the nuclear energy released from the nuclear reaction. KE of daughter nucleus is $\frac{NE}{3}$. Find N.

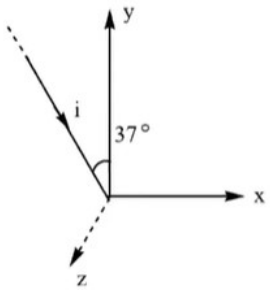
20. A longitudinal harmonic wave is propagated along a solid rod of length L whose cross-sectional area $A(x)$ varies exponentially as: $A(x)=A_0e^{-\lambda x}$ Where A_0 is the area at the input end $(x=0)$ and λ is a positive constant. As the wave travels along the rod, the average power transmitted across any cross-section remains constant. Let the displacement amplitude at the input end $(x=0)$ be s_0 and the displacement amplitude at a distance x be $s(x)$. If the ratio of the displacement amplitude at a specific position x_1 to the initial displacement amplitude is $\frac{s(x_1)}{s_0}=e^2$, find the value of the product λx_1.

21. A soldier has to cross from 80m wide river on a ropeway by grabbing on a pulley. The length of ropeway is 100m, its ends are fixed at the same height on both banks of the river and the rope is tight as shown. The soldier starts from a platform under one of the fixed points with zero initial velocity. Friction is absent everywhere. If the highest speed of the solider using the crossing is $v(in\ m/s)$. Then $\frac{v^2}{60}$ is

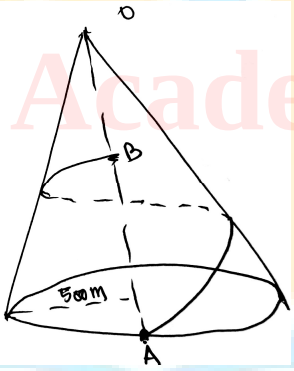


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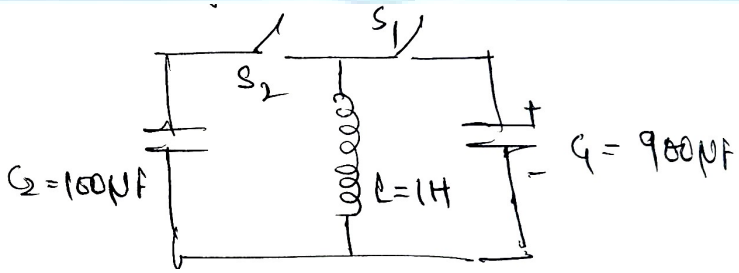
22. A direct current flows along a semi-infinite wire given by a line y=$\frac{-4}{3}x$ in x-y plane towards origin as shown in the figure. Upon reaching the origin it spreads radially all over an infinite uniform conducting plane coinciding with XZ plane. The magnetic field at the point $(R,-R,R)$ due to the current that spreads radially on the finite conducting plane mentioned above is $\frac{\mu_0 i}{4\pi R}\left(\frac{1}{\sqrt{\alpha}}-\frac{1}{\sqrt{\beta}}\right)$. Find $\alpha+\beta$



23. A mountain is of approximately conical shape of radius 500 m and slant 2km.A toy train track is to be designed between two points A and B. B is a point on the line joining apex O and point A at the base of the mountain as shown in the figure. It is given that AB=500m. The shortest possible length (in metre) of the rail track such that it revolves around the mountain completely is $500\times N$. Find N



24. In the L-C circuit, the capacitor C_1 is charged to a potential of 1 volt and the capacitor C_2 and the inductor were initially uncharged as shown in figure :



Now switch S_1 is closed at t=0 and at $t=\frac{1.5\pi}{100}$sec, switch S_2 is closed and S_1 is opened. After $\frac{0.5\pi}{100}$sec, S_2 is also opened. Find the potential difference across the capacitor C_2 in volts.

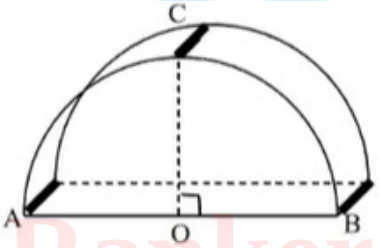
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SECTION - 2 (Maximum Marks : 24)

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Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

25. Two half disks are connected by very small insulating rods of equal lengths and cross section at three locations A,B and C. O is the centre and CO is perpendicular to AB. Now, the plates are given charge per unit area on the surface. The Young's modulus Y_a, Y_b and Y_c of rods at A,B and C are is way that the plates remains parallel after charging. Which of the following options is/are correct?



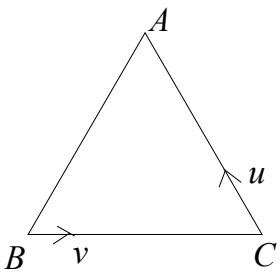
A) All three rods are under compressive stress

B) All three rods have same stress.

C) All three rods have same strain.

D) $8Y_a = 8Y_b = (3\pi - 4)Y_C$.

26. On the two vertices B, C of an equilateral triangle ABC of side d on a very rough horizontal surface, two ants of negligible dimensions start in such a way that ant from C always moves with constant velocity u directed towards A and ant from B moves at constant speed v always directed towards the other ant. They were found to meet at vertex A in time Δt from the start. The ant starting from B has a charge q and there exists a constant and uniform vertical magnetic field B_0 in the region. The other ant is neutral. Assume no slipping between ants and floor and ignore radiation effects due to accelerating charge. Also assume that there was no sudden change in velocity of ants at the start choose the correct statement(s).



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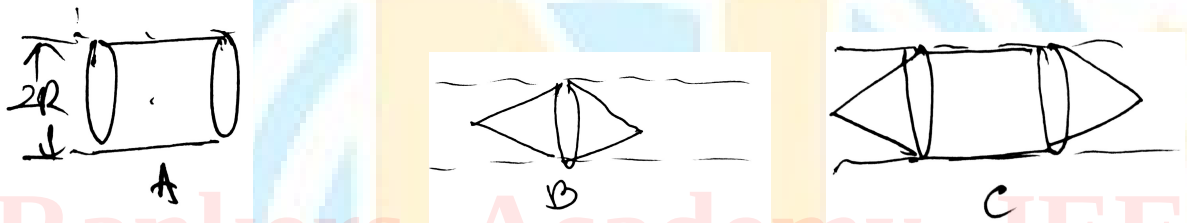
A) The value of Δt is $\frac{d}{u}$.

B) The distance travelled by ant starting from B is $\sqrt{3}d$.

C) The magnitude of average magnetic force on the ant starting from B during its journey is T_0.

D) The initial acceleration of the ant starting from B is horizontal and perpendicular to the side BC at the triangle.

27. You have two identical uniform solid right circles cones of radius R and a uniform solid cylinder of same radius R (call it object A). You make two other objects as follows with the above. Firstly, you join two cones rigidly at their common base and make a double cone (call it object B). Then you attach each one to either side of the cylinder to share the common base and axes (call it object C). It these objects A,B,C were to race down a very rough incline through same distance starting from rest. We need to rank the object in increasing order of time. Choose the correct option.



A) $B < A < C$

B) $A = B = C$

C) $B < C < A$

D) We need mass ratio of the cylinder and each cone to rank these

28. A single electron orbits around a stationary nucleus of charge $+Ze$ in a hydrogen-like atom, where Z is the atomic number and e is the magnitude of the charge on an electron.it requires $47.25\ eV$ to excite the electron from second Bohr orbit to the third Bohr orbit. Ionization energy of hydrogen atom is 13.6 Ev. Then choose correct statement (s):

A) the value of Z is 5.

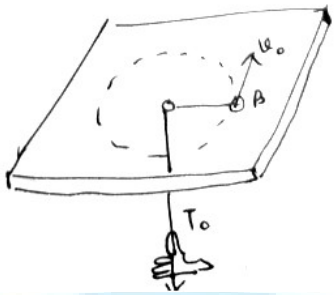
B) the energy required to excite the electron from the 3rd orbit to the 4th orbit is 16.53eV (nearly).

C) the wavelength of electromagnetic radiation required to liberate the electron completely when in the first Bohr orbit is 36.56Å.

D) the angular momentum of an electron in the second Bohr orbit is 1.056×10^{-33} Js.

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29. A small block B of mass m=0.25kg, lying on a frictionless horizontal table is all attached to a massless cord (breaking strength 40N) passing through a narrow hole C at the center of the table. Initially when the block is revolving in a circle of radius $r_o = 0.80m$ about a vertical axis through the hole, with a tangential speed of $V_o = 4m/s$. The tension in the string is T_0. and the kinetic energy of block is K_0. the string is then pulled dozen slowly from below, decreasing the radius of circular path from r_p to r so that the kinetic energy of the block is now K and the tension in the string is T. choose the correct option(s).



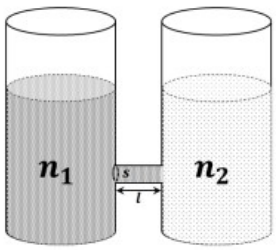
A) $T = T_o \frac{r_0^4}{r^4}$

B) $K = K_o \frac{r_0^2}{r^2}$

C) Radius r of the circulars path just when the string breaks is $0.40m$

D) Work done by the tension in the string in reducing the radius of circular from r_o to $\frac{r_o}{2}$ is 4K.

30. As shown schematically in the figure, two vessels contain water solutions (at temperature T) of potassium permanganate $(KMnO_4)$ of different concentrations n_1 and n_2 $(n_1 > n_2)$ molecules per unit volume with $\Delta n = n_1 - n_2 \ll n_1$. When they are connected by a tube of small length l and cross-sectional area S, $KMnO_4$ starts to diffuse from the left to the right vessel through the tube. Consider the collection of molecules to behave as dilute ideal gases and the difference in their partial pressure in the two vessels causing the diffusion. The speed v of the molecules is limited by the viscous force $-\beta v$ on each molecule, where β is a constant. Neglecting all terms of the order Δn^2, which of the following is/are correct? (k_B is the Boltzmann constant)



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A) The force causing the molecules to move across the tube is $\Delta n k_B T S$

B) Force balance implies $n_1 \beta v l = \Delta n k_B T$

C) Total number of molecules going across the tube per sec is $\left(\frac{\Delta n}{l}\right)\left(\frac{k_B T}{\beta}\right) S$

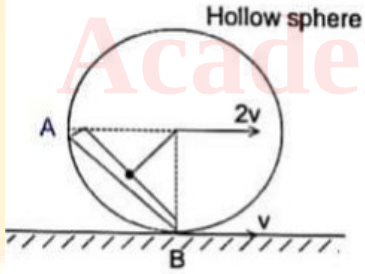
D) Rate of molecules getting transferred through the tube does not change with time

SECTION - 3 (Maximum Marks : 24)

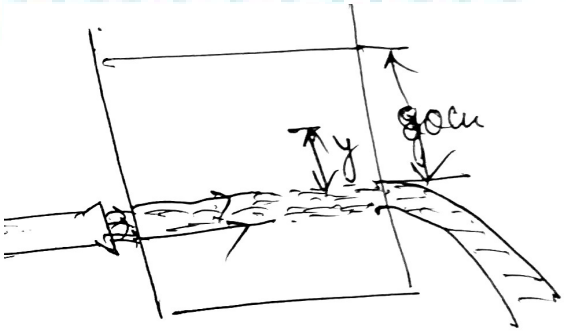
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31. A uniform rod AB of mass M and length $R\sqrt{2}$ is moving in a vertical plane inside a hollow sphere of radii R. The sphere is rolling on a fixed horizontal surface without slipping with velocity of its centre of mass 2v. When the end B is at lowest position, its speed is found to be v as shown in the figure. If the KE at red at this instant is $\frac{4mv^2}{k}$.

Find k



32. A tank of water is full upto 80 cm above the hole. The radius of tank is 1m and radius of hole is 1mm. A circular light beam of radius $1mm$ which covers the hole completely is incident on the hole from inside water. It is seen that the beam is travelling along the water stream like an optical force. At what level of water y (in mm) will the light beam begins to emerge from the curved surface of water stream? $n_w = \frac{4}{3}$



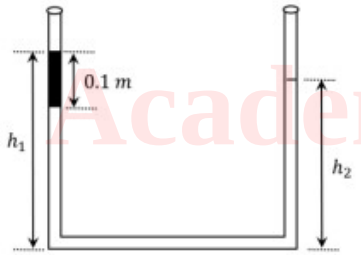
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33. A uniform rod of mass m and length l is kept at rest in horizontal position by providing supports from fingers kept at end points of rods. Keeping one finger at rest other is slide towards centre until the rod comes to the verge of motion. Then the other finger is slid towards the centre until the rod comes to the verge of motion. The process is repeated until both fingers come to the centre of the rod eventually.

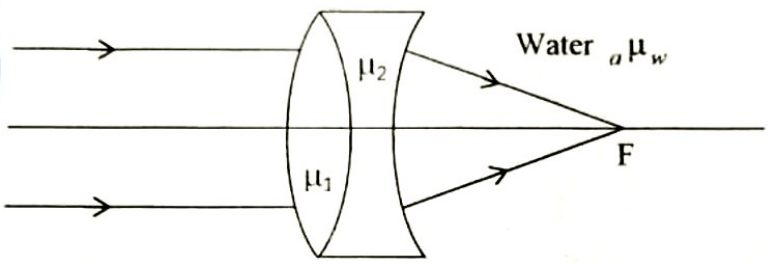
The total work done by the friction during the process is $\frac{-\mu mgl}{6}\ln X$ .Find X. Coefficient of friction between the rod and the finger are given by $\mu_k = \mu/3$ *and* $\mu_s = \mu$

34. Two capacitors with capacitance values $C_1 = 2000 \pm 10\,pF$ and $C_2 = 3000 \pm 15\,pF$ are connected in series. The voltage applied across this combination is $V = 5.00 \pm 0.02V$. The percentage error in the calculation of the energy stored in this combination of capacitors is

35. An open-ended U-tube of uniform cross-sectional area contains water $\left(density\ 10^3 kg\,m^{-3}\right)$. Initially the water level stands at 0.29 m from the bottom in each arm. Kerosene oil (a water-immiscible liquid) of density $800kg\ m^{-3}$ is added to the left arm until its length is 0.1 m, as shown in the schematic figure below. The ratio $\left(\frac{h_1}{h_2}\right)$ of the heights of the liquid in the two arms is



36. A thin equi-convex lens of flint glass (refractive index μ_1) is kept coaxially in contact with another thin equi-concave lens of crown glass (refractive index μ_2). The system is completely immersed in water $\left({}_a\mu_w = \frac{4}{3}\right)$.



Parallel rays of light incident parallel to the principal axis in water are focused by this system at a distance of 24 cm beyond the system. The thickness of the system is negligible. If the radius of curvature of the surface R = 20 cm, the difference $(\mu_1 - \mu_2)$ is

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CHEMISTRY **MAX.MARKS: 66**

SECTION – 1 (Maximum Marks: 18)

This section contains SIX (06) questions.
The answer to each question is **A SINGLE DIGIT INTEGER** rangeing from **0 TO 9 , BOTH INCLUSIVE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric
keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

37. An electron is continuously accelerated in vacuum tube by applying potential difference. If its de-Broglie wavelength is decreased by 1% then the kinetic energy of the electron is increased by x% (approximately). What is the value of x?

38. What is the oxidation number of Fe in violet coloured complex $Na_4\left[Fe(CN)_5(NOS)\right]$?

39. Identify the total number of correct statement regarding halogens and their compounds:

I) Cl_2 turns, moist starch iodide paper blue.

II) Cl_2 oxidises Iodine to periodate

III) I_2O_5 is used for the quantitative estimation of carbon mono-oxide.

IV) The acidic nature of hydrogen halides in H_2O increases from HF to HI.

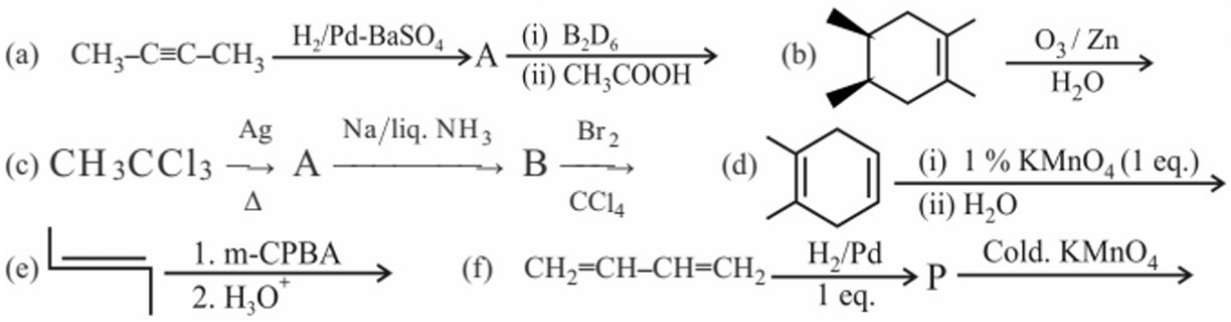
V) The general decreasing order of stability of oxides formed by Halogens $I > Br > Cl$

VI) The acidic strength of oxo-acids of chlorine increases with increase in oxidation state of chlorine.

VII) Chlorine is used as a bleaching agent in textile industry.

VIII) $Na_2S_2O_3$ is used as a antichlore in textile industries.

40. Total number of reactions in which final major product is meso:



For More Join: @IITJEE_Advanced

41. In how many cases electron gain enthalpy is negative.

i) $H(g)+e \rightarrow H^{-}(g)$

ii) $Li(g)+e \rightarrow Li^{-}(g)$

iii) $Na(g)+e \rightarrow Na^{-}(g)$

iv) $K(g)+e \rightarrow K^{-}(g)$

v) $P(g)+e \rightarrow P^{-}(g)$

vi) $O(g)+e \rightarrow O^{-}(g)$

vii) $O^{-}(g)+e \rightarrow O^{2-}(g)$

viii) $N(g)+e \rightarrow N^{-}(g)$

ix) $Ar(g)+e \rightarrow Ar^{-}(g)$

42. How many are incorrect statements?

a) CO which is a major pollutant resulting from the combustion of fuels in automobiles plays a major role in photochemical smog.

b) Classical smog has an oxidizing character while the photochemical smog is reducing in character.

c) Photochemical smog occurs in day time whereas the classical smog occurs in the morning hours.

d) During formation of smog the level of ozone in the atmosphere goes down.

e) Classical smog is good for health but not photochemical smog.

f) Chlorofluorocarbons are responsible for ozone layer depletion.

g) Green house effect is responsible for global warming.

h) Ozone layer does not permit infrared radiation from the sum to reach the earth.

i) Acid rain is mostly because of oxides of nitrogen and Sulphur.

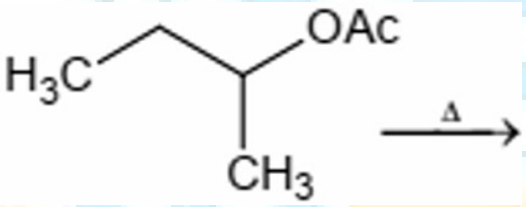
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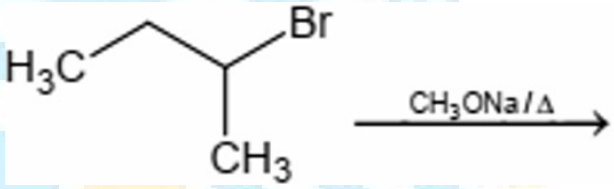
SECTION - 2 (Maximum Marks : 24)

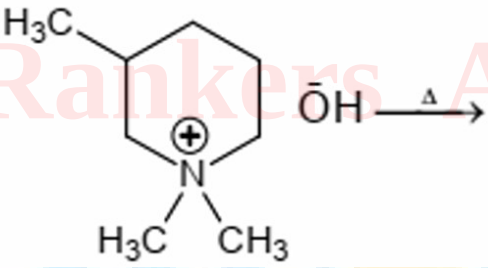
This section contains SIX (06) questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

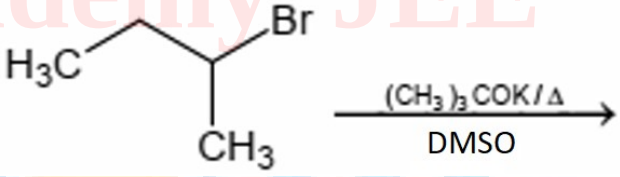
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

43. In which of the following reaction(s) product (major) formation takes place by Hofmann rule?

A) 

B) 

C) 

D) 

44. Nitrogen (I) oxide is produced by

A) thermal decomposition of ammonium nitrate

B) disproportionation of N_2O_4

C) thermal decomposition of ammonium nitrite

D) reaction of hydroxyl amine with nitrous acid

45. Select the false statement(s):

A) Brownian motion and Tyndall effect are shown by true solutions

B) Lyophobic sols don't need stabilizing agents for their preservation.

C) A mixture of noble gases can be separated by adsorption on coconut charcoal at different temperatures.

D) Higher is the gold number greater will be the protective power of a lyophilic colloid

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46. Identify the correct statement(s) regarding boron containing product (X), (Y) from the given reaction:

Colemanite $+Na_2CO_3 \rightarrow (X)+CaCO_3+(Z)$

$(Z) \xrightarrow{CO_2} (X)+Na_2CO_3$

$(X) \xrightarrow{\Delta} (Y)+(Z)$

A) On heating (Y) with copper sulphate in a non-luminous flame, cuprous metaborate is obtained.

B) Compound (X) contains tetra nuclear units.

C) The aqueous solution of (X) acts as an buffer.

D) All boron atoms in (X) are tetrahedral in shape.

47. Following two equilibria are simultaneously established in a container

$PCl_5(g) \rightleftharpoons PCl_3(g)+Cl_2; CO(g)+Cl_2(g) \rightleftharpoons COCl_2(g)$

If some $Ni(s)$ is introduced (Ni reacts with CO to form tetra carbonyl nickel) in the container establishing another equilibria.

Which among the following statements is/are definitely true about this system?

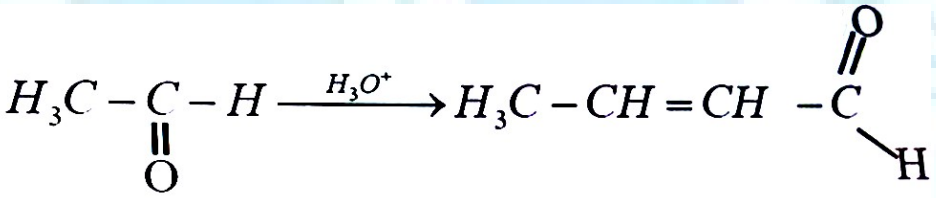
A) $COCl_2$ concentration decreases as the new equilibrium is established.

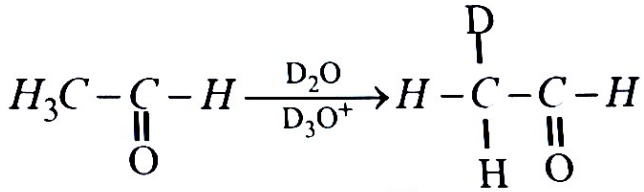
B) As Ni is a solid it will not have any impact on the existing two equilibrium reactions.

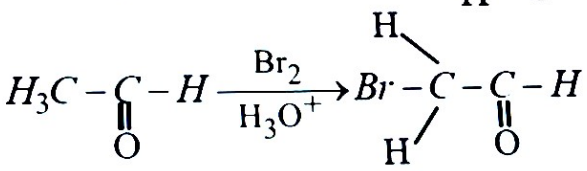
C) PCl_3 concentration decreases to re-attain the equilibrium.

D) PCl_5 concentration increases to attain the new equilibrium.

48.

Reaction I : 

Reaction II : 

Reaction III : 

Which of the following statement(s) are true regarding above reactions?

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A) Rate determining step of all three reactions follow second order kinetics

B) Rate determining step of all three reactions follow first order kinetics

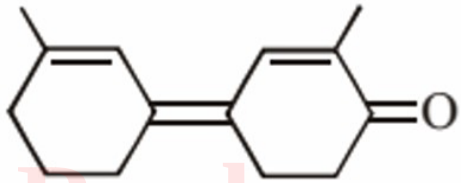
C) In reaction III, rate of reaction is independent on the initial concentration of Br_2

D) In reaction III, rate of reaction is dependent on the initial concentration of Br_2

SECTION - 3 (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

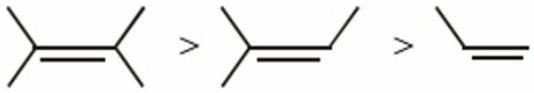
49. How many hydrogen (H) will replaced by deuterium (D) in the enol form of compound (M) when treated with $\overset{\Theta}{O}D/D_2O$ for prolonged time :

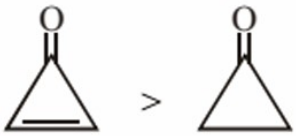


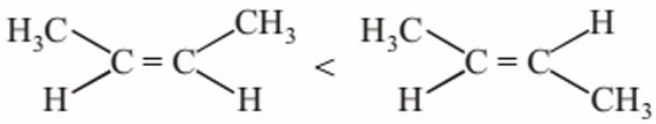
(M)

50. During disproportionation of P_4 into PH_3 and $H_2PO_2^-$ in basic medium, the mass percentage of phosphorous which is converted into PH_3 is

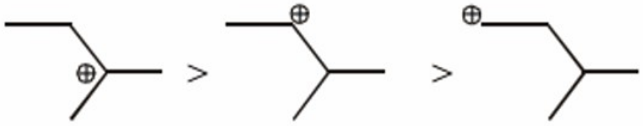
51. How many of the following orders are correctly matched?

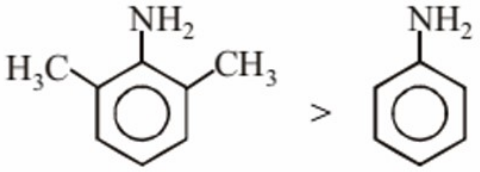
(i)  (Magnitude of heat of hydrogenation)

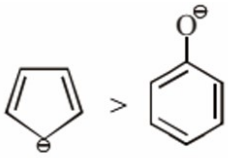
(ii)  (Solubility in water)

(iii)  (Melting point)

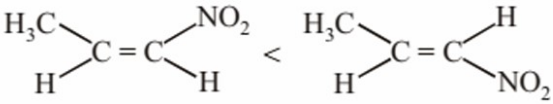
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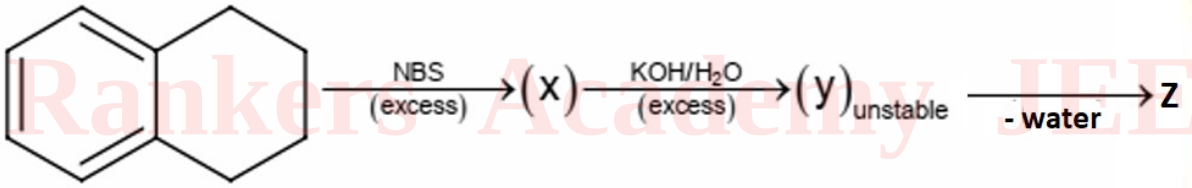
(iv)  (Carbocation stability)

(v)  (Basic strength)

(vi)  (Number of resonating structures involving in monoanion)

(vii) $-NO_2 > -CN > -F > -I$ (–I effect)

(viii)  (Dipole moment)

52. 

The total number of sp^2 – hybridized atoms present in one molecule of the end product 'Z' is 'a'. And the total number of enolizable hydrogen atoms in 'Z' is 'b'. What is the value of $a+b$?

53. A compound AB has a rock salt structure with $A:B=1:1$. The formula weight of AB is 6.023 Y amu. And the closest distance between A and B is $Y^{1/3} nm$. Then the density of the lattice in kg/m^3 is_____

54. The conductivity of a 0.0306 M solution of a weak base B is $1.53\times10^{-4}\ Scm^{-1}$. If the sum of the limiting ionic conductances for BH^+ and OH^- is $250\ Scm^2\ mol^{-1}$, the value of the base constant K_b for the base B is a $\times10^{-6}$ (in scientific notation), find the value of 'a'? (Assume $\alpha << 1$)

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